Homomorphisms, Isomorphisms and Permutation Groups

From the class book: Chapter 10, Sections 10.1 to 10.5 and Chapter 7, Sections 7.1 to 7.4

1. Key definitions

Homomorphism.

- A map $\phi : G \to G'$ with $\phi(ab) = \phi(a)\phi(b)$.
- Kernel: $\ker \phi = \{x : \phi(x) = e'\}$.
- Onto: epimorphism; one-one and onto: isomorphism ($G \cong G'$); $G \to G$: endomorphism; isomorphism $G \to G$: automorphism.

Automorphism groups.

- $\mathrm{Aut}(G)$ is the group of automorphisms under composition.
- For $a \in G$, $\phi_a(x) = axa^{-1}$ is an inner automorphism; these form $\mathrm{Inn}(G) \trianglelefteq \mathrm{Aut}(G)$.

Symmetric group.

- S_n is the group of bijections of $\{1, \ldots, n\}$, of order $n!$.
- Products are composed right to left: $\sigma\tau$ means "first τ, then σ"; e.g. $(123)(12) = (13)$.
- Order, cycle type and parity do not depend on the convention.

Cycles.

- An r-cycle $(a_1\, a_2\, \ldots\, a_r)$ sends $a_1 \mapsto a_2 \mapsto \cdots \mapsto a_r \mapsto a_1$ and fixes all else; a 2-cycle is a transposition.
- The cycle type of σ is the partition of n given by its disjoint cycle lengths (fixed points as 1-cycles); there are $p(n)$ cycle types.

Even/odd, alternating group.

- σ is even (odd) if it is a product of an even (odd) number of transpositions.
- The even permutations form $A_n \trianglelefteq S_n$, of index 2, $|A_n| = n!/2$ ($n \geq 2$).

2. Main results

Theorem (Properties of homomorphisms)

- $\phi(e) = e'$, $\phi(a^{-1}) = \phi(a)^{-1}$, $\phi(a^n) = \phi(a)^n$; $o(\phi(a)) \mid o(a)$.
- $\ker \phi \trianglelefteq G$; ϕ is one-one $\iff$ $\ker \phi = \{e\}$; ϕ is $|\ker \phi|$-to-one.
- Images and preimages of (normal) subgroups are (normal) subgroups (normal in $\phi(G)$ for images).
- Images of abelian (cyclic) groups are abelian (cyclic). $|\phi(G)|$ divides both $|G|$ and $|G'|$.

Theorem (Isomorphism theorems)

- (1) $G/\ker\phi \cong \phi(G)$, so $|G| = |\ker\phi|\,|\phi(G)|$.
- (2) $H \le G, N \trianglelefteq G \Rightarrow HN \le G, H\cap N \trianglelefteq H, HN/N \cong H/(H\cap N)$.
- (3) $K \subseteq H$ normal in $G \Rightarrow (G/K)/(H/K) \cong G/H$.
- (Correspondence) Subgroups of $G/N \leftrightarrow$ subgroups of G containing N, preserving normality.

Standard uses: $\det : GL(n,\mathbb{R}) \to \mathbb{R}^*$ gives $GL/SL \cong \mathbb{R}^*$; sgn gives $S_n/A_n \cong \mathbb{Z}_2$; $x \mapsto e^{2\pi i x}$ gives $\mathbb{R}/\mathbb{Z} \cong S^1$.

Theorem (Cayley)

- Every group G embeds in S_G via $g \mapsto \lambda_g$, $\lambda_g(x) = gx$; a group of order n embeds in S_n.
- *Generalized:* if $[G:H] = n$, G acts on the cosets of H, giving $G \to S_n$ with kernel inside H; if $|G| \nmid n!$ the kernel is a proper non-trivial normal subgroup.

Theorem (Automorphisms)

- $\operatorname{Inn}(G) \cong G/Z(G)$.
- G abelian $\iff$ $x \mapsto x^{-1}$ is an automorphism $\iff$ $\operatorname{Inn}(G)$ trivial.
- $\operatorname{Aut}(\mathbb{Z}_n) \cong U(n)$, of order $\varphi(n)$, cyclic iff $n = 1, 2, 4, p^k, 2p^k$.

Theorem (Counting homomorphisms)

A homomorphism from $\langle a\rangle$ is fixed by the image x of a, which can be any x with $x^{o(a)} = e$.

$$\#\mathrm{Hom}(\mathbb{Z}_m,\mathbb{Z}_n) = \gcd(m,n), \qquad \#\mathrm{Hom}(\mathbb{Z}_m,G) = \#\{x\in G : x^m = e\}, \qquad \#\mathrm{Hom}(\mathbb{Z},G) = |G|.$$

- Onto $\mathbb{Z}_m \to \mathbb{Z}_n$: $\varphi(n)$ if $n \mid m$, else 0.
- One-one $\mathbb{Z}_m \to \mathbb{Z}_n$: $\varphi(m)$ if $m \mid n$, else 0.
- If A is abelian, $\#\mathrm{Hom}(G,A) = \#\mathrm{Hom}(G/G', A)$.

Theorem (Cycle decomposition)

- Every permutation is a product of disjoint cycles, unique up to order; disjoint cycles commute.
- Order $=$ lcm of cycle lengths.
- $(a_1\cdots a_r) = (a_1a_r)(a_1a_{r-1})\cdots(a_1a_2)$ ($r-1$ transpositions) and $(a_1\cdots a_r)^{-1} = (a_r\cdots a_1)$.

Theorem (Parity)

- $\operatorname{sgn} : S_n \to \{\pm 1\}$ is a homomorphism.
- An r-cycle is even $\iff$ r is odd.
- If σ has k cycles (fixed points included), $\operatorname{sgn}\sigma = (-1)^{n-k}$; also $\operatorname{sgn}\sigma = (-1)^{\mathrm{inv}(\sigma)}$.

Theorem (Subgroups and generators of S_n)

- Every $H \leq S_n$ lies in A_n or is half even, half odd; so subgroups of odd order lie in A_n.
- S_n is generated by the transpositions, by $(12), (13), \ldots, (1n)$, or by (12) and $(12\cdots n)$.
- A_n ($n \geq 3$) is generated by 3-cycles.
- A_n is the only index-2 subgroup of S_n.
- $Z(S_n) = \{e\}$ for $n \geq 3$; $S_3 \cong D_3 \cong GL(2, \mathbb{Z}_2)$.

Theorem (Counting in S_n)

Number of r-cycles: $\binom{n}{r}(r-1)! = \dfrac{n!}{r\,(n-r)!}$. If the cycle type has a_k cycles of length k:

$$\#\{\sigma \text{ of this type}\} = \frac{n!}{\prod_k k^{a_k}\, a_k!}.$$

- Two permutations are conjugate in S_n iff they have the same cycle type.
- An S_n-class splits into two A_n-classes iff its cycle lengths are distinct and odd.

3. Tables to remember

Isomorphism invariants

Pair	Isomorphic?	Reason
$(\mathbb{Z}, +), (2\mathbb{Z}, +)$	yes	$n \mapsto 2n$
$(\mathbb{R}, +), (\mathbb{R}^+, \cdot)$	yes	$x \mapsto e^x$
$(\mathbb{Q}, +), (\mathbb{Q}^*, \cdot)$	no	-1 has order 2 in $\mathbb{Q}^*$
$(\mathbb{R}^*, \cdot), (\mathbb{C}^*, \cdot)$	no	i has order 4
$U(8), U(12), K_4$	yes	order 4, all elements of order ≤ 2
D_4, Q_8	no	5 elements of order 2 vs 1

Automorphism groups

G	$\mathbb{Z}$	$\mathbb{Z}_n$	$\mathbb{Z}_p \times \mathbb{Z}_p$	K_4	$\mathbb{Z}_2^3$	S_3	Q_8	D_4	S_n $(n \neq 2, 6)$
$\mathrm{Aut}(G)$	$\mathbb{Z}_2$	$U(n)$	$GL(2, \mathbb{Z}_p)$	S_3	$GL(3, \mathbb{Z}_2)$	S_3	S_4	D_4	S_n
order	2	$\varphi(n)$	$(p^2-1)(p^2-p)$	6	168	6	24	8	$n!$

Homomorphism counts

From $\to$ to	all	onto	one-one
$\mathbb{Z}_6 \to \mathbb{Z}_8$	2	0	0
$\mathbb{Z}_{16} \to \mathbb{Z}_8$	8	4	0
$\mathbb{Z} \to \mathbb{Z}_n$	n	$\varphi(n)$	0
$S_n \to \mathbb{Z}_m$ $(n \geq 2)$	1 (m odd), 2 (m even)	1 if $m \leq 2$, else 0	–
$A_4 \to \mathbb{Z}_m$	3 if $3 \mid m$, else 1	2 if $m = 3$	–
$Q_8 \to K_4$	16	6	0
$K_4 \to Q_8$	4	0	0
$K_4 \to S_4$	52	0	24

- $\#\mathrm{Hom}(K_4, G) = 1 + 3t + 6k$, $t =$ number of elements of order 2, $k =$ number of Klein four-subgroups ($6k$ of them one-one).
- For S_4: $t = 9, k = 4$, giving 52.

Element orders in S_n and A_n

n	1	2	3	4	5	6	7	8	9	10
$p(n)$	1	2	3	5	7	11	15	22	30	42
max order in S_n	1	2	3	4	6	6	12	15	20	30

Order	1	2	3	4	5	6	7	10	12
S_4	1	9	8	6	–	–	–	–	–
A_4	1	3	8	–	–	–	–	–	–
S_5	1	25	20	30	24	20	–	–	–
A_5	1	15	20	–	24	–	–	–	–
S_6	1	75	80	180	144	240	–	–	–
S_7	1	231	350	840	504	1470	720	504	420

Cycle types of S_4

Type	Example	Number	Order, parity
$(1,1,1,1)$	e	1	1, even
$(2,1,1)$	(12)	6	2, odd
$(2,2)$	$(12)(34)$	3	2, even
$(3,1)$	(123)	8	3, even
(4)	(1234)	6	4, odd

4. Exam shortcuts and standard counter-examples

Exam shortcut.

- Non-isomorphism: compare order, abelianness, centre, and number of elements of each order (especially order 2 and 4). One mismatch suffices.
- Equal counts of element orders do not force isomorphism in general (smallest counter-examples have order 16).

Homomorphisms into abelian groups.

- Factor through G/G': $S_n/S_n' \cong \mathbb{Z}_2$, $A_4/A_4' \cong \mathbb{Z}_3$, $Q_8/Q_8' \cong K_4$.
- Onto maps $G \to H$: (normal N with $G/N \cong H$) $\times |\mathrm{Aut}(H)|$.
- No non-trivial homomorphism from a finite group to $\mathbb{Z}$; $\mathrm{Hom}(\mathbb{Z}, \mathbb{Z})$ is infinite with exactly 2 onto maps.

Traps.

- $\mathrm{Aut}(\mathbb{Z}_n) \cong U(n)$, not $\mathbb{Z}_n$; $\mathrm{Aut}(\mathbb{Z}_{15}) \cong \mathbb{Z}_2 \times \mathbb{Z}_4$ is not cyclic.
- $|\mathrm{Inn}(G)| = [G : Z(G)]$: $|\mathrm{Inn}(S_3)| = 6$, $|\mathrm{Inn}(D_4)| = |\mathrm{Inn}(Q_8)| = 4$.
- Every abelian group of odd order > 1 has the non-trivial automorphism $x \mapsto x^{-1}$.

Permutation shortcuts.

- Parity: (symbols moved) − (number of non-trivial cycles) even $\Rightarrow$ even.
- Elements of order 2 in S_n: $\sum_{k \geq 1} \frac{n!}{2^k k!(n-2k)!}$ (in S_5: $10 + 15 = 25$; in S_4: $6 + 3 = 9$, not just the transpositions).
- Possible orders in S_n are the lcm's of partitions of n; the maximum is not n (6 in S_5, 12 in S_7).
- Non-disjoint cycles must be multiplied out: $(12)(23)$ has order 3.

5. Solved examples

Example 1. Find the number of homomorphisms $\mathbb{Z}_{12} \to \mathbb{Z}_{18}$, and the numbers of onto homomorphisms $\mathbb{Z}_{12} \to \mathbb{Z}_6$ and one-one homomorphisms $\mathbb{Z}_6 \to \mathbb{Z}_{18}$.

Solution

- $\#\mathrm{Hom}(\mathbb{Z}_{12}, \mathbb{Z}_{18}) = \gcd(12, 18) = 6$.
- Onto $\mathbb{Z}_{12} \to \mathbb{Z}_6$: $6 \mid 12$, so $\varphi(6) = 2$ ($\bar{1} \mapsto \bar{1}$ or $\bar{5}$).
- One-one $\mathbb{Z}_6 \to \mathbb{Z}_{18}$: $6 \mid 18$, so $\varphi(6) = 2$ ($\bar{1} \mapsto \bar{3}$ or $\overline{15}$). **Answers: 6, 2, 2.**

Example 2. Find $\#\mathrm{Hom}(\mathbb{Z}_4, S_3)$, $\#\mathrm{Hom}(S_3, \mathbb{Z}_4)$ and $\#\mathrm{Hom}(\mathbb{Z}_6, S_4)$.

Solution

- $\mathbb{Z}_4 \to S_3$: count $x \in S_3$ with $x^4 = e$, i.e. $o(x) \in \{1, 2\}$: $1 + 3 = 4$.
- $S_3 \to \mathbb{Z}_4$: $\mathbb{Z}_4$ is abelian, so this equals $\#\mathrm{Hom}(S_3/A_3, \mathbb{Z}_4) = \#\mathrm{Hom}(\mathbb{Z}_2, \mathbb{Z}_4) = \gcd(2, 4) = 2$.
- $\mathbb{Z}_6 \to S_4$: $x^6 = e \iff o(x) \in \{1, 2, 3, 6\}$; S_4 has $1 + 9 + 8 + 0 = 18$ such elements. **Answers: 4, 2, 18.**

Example 3. Composing right to left, write $\sigma = (1\,2\,3)(2\,4\,5)$ as a product of disjoint cycles; find its order and parity.

Solution

- Apply (245) first: $1 \to 1 \to 2$; $2 \to 4 \to 4$; $4 \to 5 \to 5$; $5 \to 2 \to 3$; $3 \to 3 \to 1$.
- So $\sigma = (1\,2\,4\,5\,3)$, a 5-cycle: order 5, even (as it must be: a product of two 3-cycles is even).

Example 4. Find the number of elements of order 10 in S_7 and of order 6 in S_6.

Solution

- Order 10 in S_7 needs a 5-cycle and a 2-cycle: type $(5, 2)$, count $\dfrac{7!}{5 \cdot 2} = 504$.
- Order 6 in S_6: types (6) and $(3, 2, 1)$: $\dfrac{6!}{6} + \dfrac{6!}{3 \cdot 2 \cdot 1} = 120 + 120 = 240$.
- (Type $(3, 3)$ has order 3 and $(2, 2, 2)$ order 2.) **Answers: 504, 240.**

Example 5. Which of $U(16)$, $\mathbb{Z}_8$, $\mathbb{Z}_4 \times \mathbb{Z}_2$, $\mathbb{Z}_2^3$ are isomorphic?

Solution

- In $U(16) = \{1, 3, 5, 7, 9, 11, 13, 15\}$: $7^2 = 49 \equiv 1$, $9^2 = 81 \equiv 1$, $15^2 \equiv 1$, while $3, 5, 11, 13$ have order 4.
- So 3 elements of order 2, 4 of order 4.
- $\mathbb{Z}_8$ has 1 element of order 2; $\mathbb{Z}_2^3$ has 7; $\mathbb{Z}_4 \times \mathbb{Z}_2$ has 3 of order 2 and 4 of order 4.
- Hence $U(16) \cong \mathbb{Z}_4 \times \mathbb{Z}_2$, and no other pair is isomorphic.

Example 6. Find $|\mathrm{Aut}(\mathbb{Z}_{20})|$, whether it is cyclic, and $|\mathrm{Aut}(\mathbb{Z}_3 \times \mathbb{Z}_3)|$.

Solution

- $\mathrm{Aut}(\mathbb{Z}_{20}) \cong U(20)$, of order $\varphi(20) = 8$.
- Element orders in $U(20)$: 1 (one), 2 ($9, 11, 19$), 4 ($3, 7, 13, 17$); no element of order 8, so $U(20) \cong \mathbb{Z}_2 \times \mathbb{Z}_4$, not cyclic (consistent with $20 \neq p^k, 2p^k$).
- $\mathrm{Aut}(\mathbb{Z}_3 \times \mathbb{Z}_3) \cong GL(2, \mathbb{Z}_3)$, of order $(9 - 1)(9 - 3) = 48$.